# RiskRadar

From raw news to risk signals — an AI/NLP Risk Engine that rebalances an index and stress-tests a bank portfolio, live.

`2+ sources · news + social` `3 signals · sentiment / event / impact` `BOTH modules · rebalancing + stress testing`

`0.3 ms per signal · fully offline`

Phase III — Case Study Competition · Individual submission
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# Markets move on news. Risk systems only eat numbers.

The case study asks for an engine that converts real-time unstructured text into structured risk signals — then at least one module that acts on them. I built the core **and both** modules.

CORE (REQUIRED)

## AI/NLP Risk Engine

Ingests **financial news + social posts** (2 source types; live NewsAPI/GDELT/Reddit connectors optional, fail-soft). Every item → **sentiment (–1…+1)**, **event class (12 types)**, **impact (1–10)** — each carrying its own evidence.

MODULE A (BUILT)

## Index Rebalancing

Subscribes to sentiment; tilts a 20-stock mock S&P-100 index with w×(1+0.25·s), 2–12% band, full audit trail — weights drift live on the dashboard.

MODULE B (BUILT)

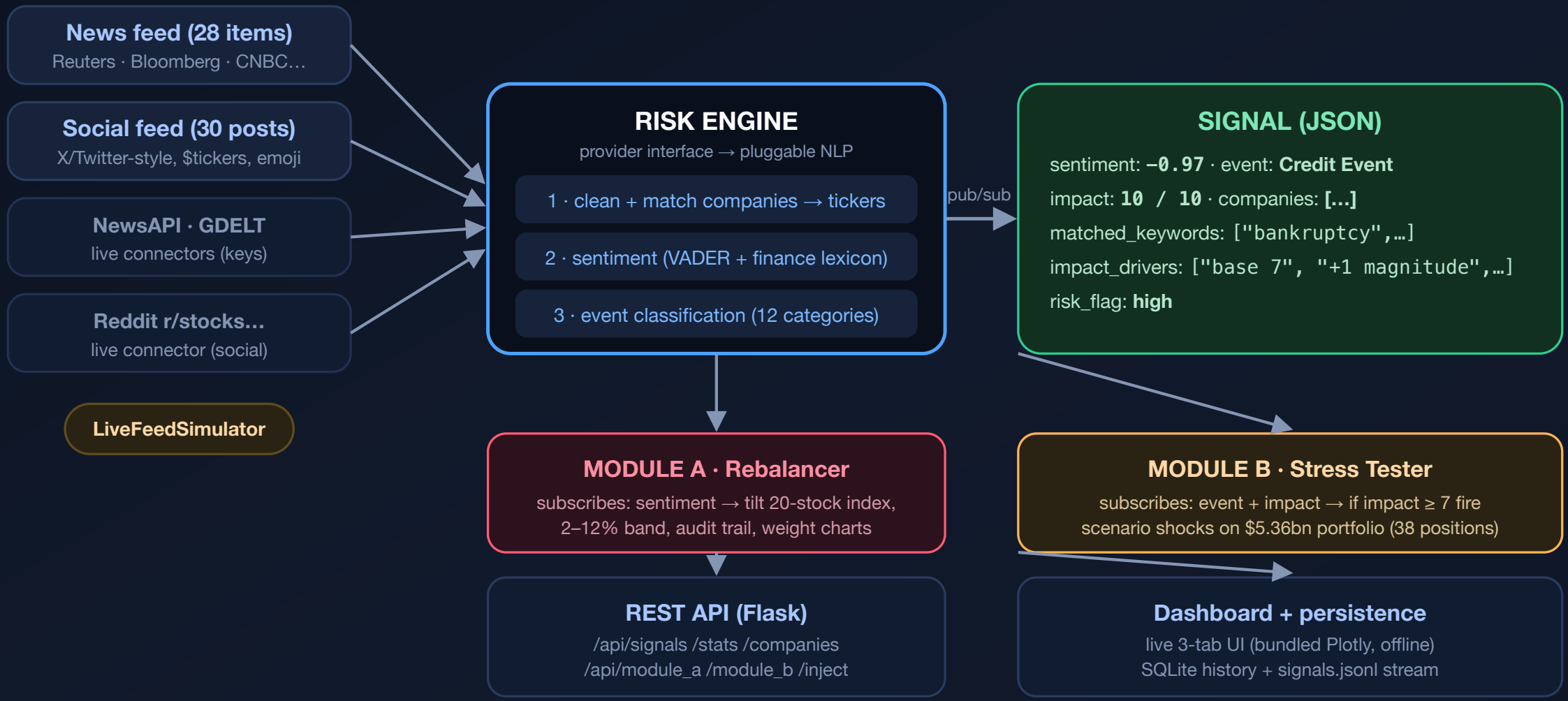
## Stress Testing

Subscribes to event + impact; any **impact ≥ 7** event fires a scenario shock (e.g. rates +200bp, equity −10%) on a **$5.36bn** wholesale portfolio — 38 positions, 9 asset classes — with before/after P&L.

> "One engine, three explainable numbers, two consumers — tactical rebalancing and strategic stress testing — running live, offline, on a laptop."

# RiskRadar

ARCHITECTURE · ONE PROCESS, ONE DATA-FLOW DIRECTION



# RiskRadar

## IMPLEMENTATION HIGHLIGHTS · EVERY TEXT → THREE NUMBERS + EVIDENCE

**sentiment –1…+1 · VADER + 150 finance terms**

**event · 12 classes, priority rules**

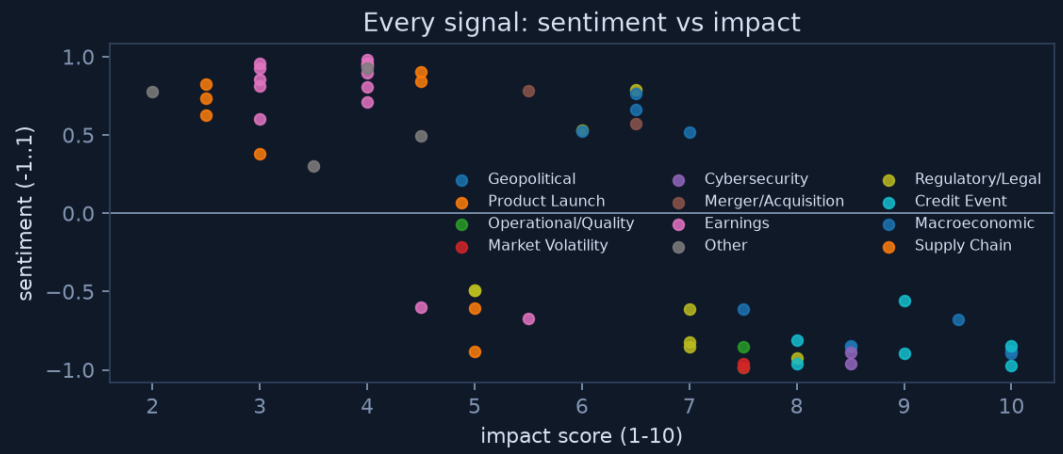
**impact 1–10 · transparent formula**

```
IN : "OPEC+ surprise production cut escalates
       trade tensions with Washington"
OUT: { "sentiment": -0.84,
       "event_type": "Geopolitical",
       "impact": 8.5,  "risk_flag": "high",
       "matched_keywords": ["opec","escalating",…],   // why this
class
       "impact_drivers": ["base 6.5 (Geo)",
                          "+1 urgency"," +1 negative"] }          // why
8.5
```

93% of 58 corpus items got a meaningful class; 54/58 · 0.3 ms/signal · ~3,300/sec on one core.

Every signal carries its evidence — keywords + drivers — because a risk number that can't answer "why?" is worthless.

```
demo killer move: type any unseen headline in the dashboard →
verdict in 0.3 ms
```




every signal: sentiment × impact, colored by event class — the red zone is where Module B wakes up

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MODULE A · TACTICAL — SENTIMENT → INDEX WEIGHTS, LIVE

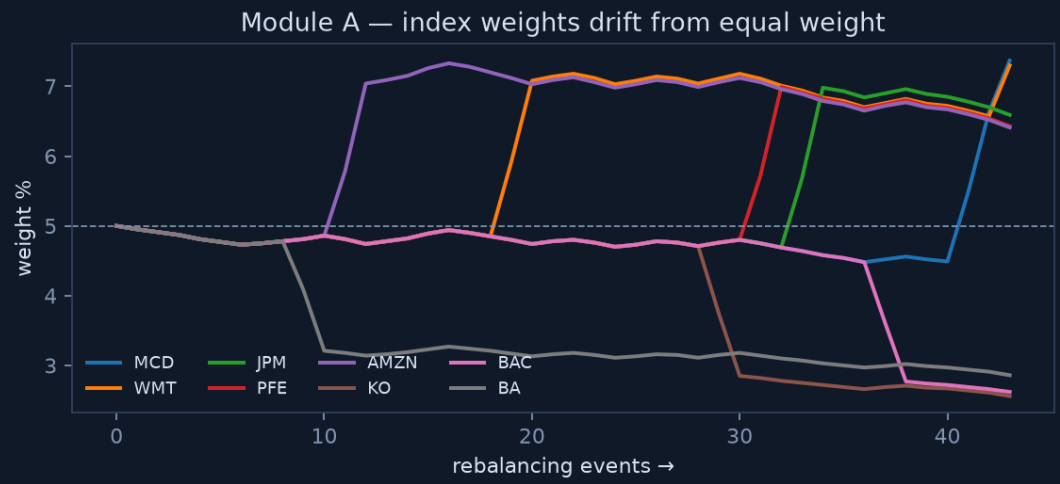
```
w_new = w_old × (1 + 0.25 × sentiment)
w     = clamp(w, 2%, 12%)      // concentration limits
w    /= Σ w                    // renormalize to 100%
```

- **20 stocks** (S&P 100 subset), start equal-weighted at 5%
- **45 weight changes** logged in the demo run, each with the signal that caused it — full audit trail
- Positive sentiment earns share; negative sheds it. Compounds across the stream — that's the drift you see below.

WINNERS
**MCD 7.4%**
**WMT 7.3%**
**JPM 6.6%**

LOSERS
**KO 2.6%**
**BAC 2.6%**
**BA 2.9%**

WHY
**recall −0.98 · subprime warnings · supply-chain hit**




weights drifting from the 5% equal-weight line as sentiment arrives

# RiskRadar

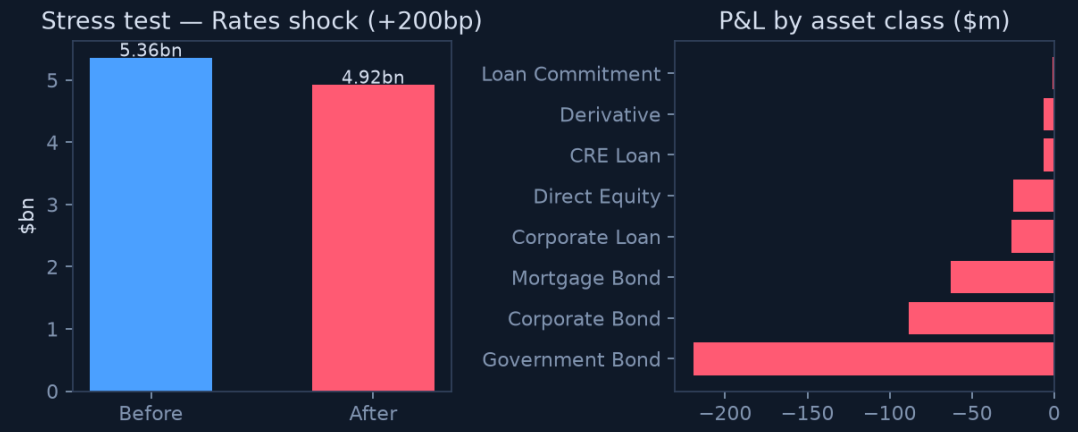
MODULE B · STRATEGIC — HIGH-IMPACT EVENTS FIRE STRESS SCENARIOS

- **$5.36bn synthetic wholesale portfolio** — 38 positions, 9 asset classes: corporate loans $1.6bn, CRE, govt bonds, IG/HY credit, MBS, derivatives (swaps, CDS), commitments, equity.
- **Trigger: impact ≥ 7** (the case study's own threshold) → event-typed shock:
  `rates +200bp` `spreads +250bp` `equity −10%`
- First-order, fully traceable math: bonds lose `notional × duration × Δrate` — the 30Y Treasury alone bleeds **$92.5m** in the rates shock.

RATES SHOCK
**−8.1%**
**−$435m**
Macroeconomic · impact 9

CREDIT SHOCK
**−6.6%**
**−$355m**
Chapter 11 · impact 10

GEO SHOCK
**−4.0%**
**−$215m**
OPEC cut · impact 8.5




before/after valuation + P&L by asset class — you can see exactly where the loss lands

# RiskRadar

DOMAIN IMPACT · CHALLENGES · HONEST LIMITS · WHAT'S NEXT

### HARDEST PROBLEMS WE SOLVED

- Unreliable live APIs → **fail-soft architecture** + live-feed simulator; demo can't break
- VADER didn't know finance → **150-term lexicon**; classification conflicts → priority rules
- "Impact 7" had no definition → **transparent formula**, drivers stored on every signal

### HONEST LIMITATIONS

- Sentiment is lexicon-based (v1), not learned — **chosen for explainability**, not hidden power
- Impact bases are domain-judged, not back-calibrated on history
- Stress math is first-order — no VaR, correlations, netting

### ONE MORE WEEK →

- **FinBERT** behind the provider interface (one-file swap)
- Backtest Module A on the case study's Kaggle tweet/price datasets
- Scenario-VaR layer for Module B · Kafka/WebSockets · auth + Docker

### DOMAIN IMPACT — WHY IT MATTERS

One pipeline feeds two desks: a **tactical desk** (rebalance as sentiment arrives) and a **strategic desk** (stress the book when high-impact events land) — the same workflow a market index team and a wholesale-risk team run separately today, at a fraction of the cost. **Efficiency vs naive approach:** 0.3 ms per signal (~3,300/sec on one laptop core) — no GPU, no API calls, no internet — and every number explains itself.

"One engine · three explainable numbers · two consumers — tactical rebalancing and strategic stress testing, running live, offline, on one laptop. Ask me anything — or throw any headline at it right now."

`python run.py` `dashboard → localhost:8000` `repo: RiskRadar/` `report: Project Report & Q&A PDF`